

Setup and Deployment Guide

For VS Code and GitHub (Desktop or the VS Code Git integration), plus Power BI and Tableau Public

Compliance Audit Data Automation | Rahul Mistari

What you will need before starting

Python 3.9 or later, installed and on your system PATH.

VS Code, with the Python extension installed (Extensions panel, search "Python", the Microsoft one).

Git and a GitHub account. If you already have GitHub Desktop or GitHub's CLI installed, either works alongside VS Code; this guide uses VS Code's own built-in Git and GitHub support, which needs nothing extra beyond signing in once.

Tableau Public Desktop (free), only if you want to publish there. Optional.

Power BI Desktop (free, Windows only), only if you want to publish there. Optional.

Terminal commands below run in VS Code's own built-in terminal (menu: Terminal, then New Terminal, or Ctrl+`), so you do not need to open a separate terminal app.

Step 1: Open the project in VS Code

If you were given a .zip file, unzip it anywhere convenient.

Open VS Code. Go to File, then Open Folder, and select the unzipped compliance-audit-data-automation folder.

VS Code will open with the project's file tree on the left: src, docs, tests, data, and so on.

Open the built-in terminal: Terminal menu, then New Terminal. It opens already pointed at the project folder, so no cd is needed.

Step 2: Run it locally

In the VS Code terminal, create and activate a virtual environment (optional but recommended), then install the required packages:

```
python -m venv .venv
source .venv/bin/activate # on Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

If VS Code shows a pop-up asking whether to use this environment for the workspace, choose Yes. This makes VS Code's Python extension aware of the installed packages, which helps with things like auto-complete and test discovery.

Generate the synthetic sample data:

```
python generate_sample_data.py
```

Run the full validation workflow. This writes the audit trail and summary to the outputs folder:

```
python -m src.main
```

Run the test suite. You can either type it in the terminal, or use VS Code's Testing panel (the flask icon in the left sidebar) after pointing it at pytest, via Command Palette (Ctrl+Shift+P) then "Python: Configure Tests":

```
pytest -v
```

Build the Tableau Public / Power BI ready export files:

```
python build_bi_exports.py
```

At this point the project is fully working, including the outputs folder, the data/bi_exports folder, and the dashboard file at docs/index.html. Right-click index.html in the VS Code file tree and choose Reveal in File Explorer (or Finder), then open it in a browser to preview the dashboard before publishing anything.

Step 3: Push the project to GitHub, from inside VS Code

Click the Source Control icon in the left sidebar (it looks like a branching line, or press Ctrl+Shift+G).

If this folder is not a Git repository yet, VS Code shows a button labeled Initialize Repository. Click it.

If the project already came with Git history (it does, if you unzipped a copy that already had a .git folder), VS Code detects that automatically and shows your changes and commit history instead; skip to the next step.

You should now see a button labeled Publish to GitHub. Click it.

VS Code will ask you to sign in to GitHub, if you are not already. Follow the prompt; it opens your browser to authorize, then returns to VS Code.

Choose a repository name (for example compliance-audit-data-automation), choose Public (so the dashboard can be publicly viewed later), and confirm.

VS Code creates the GitHub repository and pushes all the project files to it in one step. When it finishes, a notification appears with a link to open the repository on GitHub.com.

For any changes after this point: stage them in the Source Control panel (the plus icon next to each changed file, or the plus next to Changes for all of them), type a commit message in the box at the top, click the checkmark to commit, then click Sync Changes to push.

If you would rather use GitHub Desktop instead of VS Code's built-in integration: open GitHub Desktop, choose File, then Add Local Repository, select the project folder, then Publish Repository. The result is the same either way; use whichever you find more comfortable.

Step 4: Publish the dashboard with GitHub Pages

Open the repository on GitHub.com (the link from the notification in Step 3, or navigate to it from your GitHub profile).

Click Settings, then Pages in the left sidebar.

Under Build and deployment, set Source to Deploy from a branch.

Set Branch to main, and the folder to /docs, then click Save.

Wait one to two minutes. Refresh the page. GitHub will show your live URL, in the form:

<https://your-username.github.io/compliance-audit-data-automation/>

Open that URL. You should see the results dashboard, with tabs for Results, What do these rules mean, Rules by country, and Where this data comes from.

The same URL plus a filename gives direct links to the three PDF companions, for example:

https://your-username.github.io/compliance-audit-data-automation/Project_Explainer.pdf

Step 5: Add the dashboard link to your portfolio

Copy the GitHub Pages URL from Step 4.

In your portfolio site or resume, add it as a project link, for example labeled "Live dashboard" or "View results."

Optional: back in VS Code, update the placeholder link near the top of README.md with the same URL, then commit and sync, so anyone reading the repository can find it too.

Step 6: Publish to Tableau Public

Install Tableau Public Desktop (free) from Tableau's website, if you have not already.

Open it. Choose Connect, then Text File (for a .csv) or Microsoft Excel (for a .xlsx).

Select data/bi_exports/audit_results.csv or data/bi_exports/compliance_audit_workbook.xlsx from the project folder. In VS Code's file tree, right-click either file and choose Copy Path if you need the exact location.

Create a new worksheet. Drag Applies To onto Columns and Result onto Rows, then choose a bar chart. This recreates the "Findings by domain" chart from the HTML dashboard.

Create a second worksheet. Drag Result onto Color and choose a pie chart, for the pass/fail split.

Create a third worksheet as a simple table, filtered to Needs Review equals Yes, for the escalation list.

Create a new Dashboard and drag all three worksheets onto it.

Go to File, then Save to Tableau Public. Sign in, or create a free account if you do not have one.

Tableau Public publishes the workbook and gives you a public URL. Copy that URL for your portfolio.

Step 7: Publish to Power BI

Install Power BI Desktop (free, Windows only) from Microsoft's website, if you have not already.

Open it. Go to Home, then Get Data, then choose Text/CSV or Excel workbook.

Select data/bi_exports/audit_results.csv or data/bi_exports/compliance_audit_workbook.xlsx, then click Load.

Add a Card visual for total checks, a Stacked Column Chart (Applies To on the axis, Result as the legend), and a Donut Chart for the pass/fail split.

Add a Table visual filtered to Needs Review equals Yes, for the escalation list.

Go to File, then Publish, then Publish to Power BI. This requires a free Power BI account and publishes the report to a workspace in the Power BI service.

For a public link similar to Tableau Public, open the report in the Power BI service in your browser, go to File, then Publish to web. This generates a public, embeddable URL. Copy it for your portfolio.

Where each link ends up

Portfolio item	Where the link comes from
Live dashboard	GitHub Pages URL, from Step 4
Source code	The GitHub repository URL, from Step 3
Project explainer (PDF)	GitHub Pages URL plus /Project_Explainer.pdf
Why these rules exist (PDF)	GitHub Pages URL plus /Why_These_Compliance_Rules_Exist.pdf
This setup guide (PDF)	GitHub Pages URL plus /Setup_and_Deployment_Guide.pdf

Portfolio item	Where the link comes from
Tableau Public dashboard	The public URL from Step 6
Power BI report	The published-to-web URL from Step 7

Notes and troubleshooting

If VS Code's Source Control panel shows Sync Changes fail because the remote has commits you do not have locally, and you are certain your local version should replace what is on GitHub, use the terminal instead: `git push -u origin main --force`. This permanently overwrites the remote history, so only do it if that is what you intend.

Commits only show up as yours on GitHub, with your avatar and profile linked, if the commit's author email matches a verified email on your GitHub account. VS Code sets this automatically once you sign in to GitHub through it; if commits show a generic name instead, check Settings, then search "git.user" in VS Code, or run `git config user.email` in the terminal.

To refresh the sample data and results with a new random run, repeat Step 2, then stage, commit, and sync the changed files in data, outputs, and data/bi_exports from the Source Control panel.

Neither Tableau Public nor Power BI accept an uploaded dashboard file directly from outside their own applications. Both require opening the data in the desktop app and publishing from there, which is the normal workflow for each platform, not a limitation specific to this project.

Quick command reference

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python generate_sample_data.py
python -m src.main
pytest -v
python build_bi_exports.py
```

```
# GitHub, if not using VS Code's Source Control panel:
git add -A
git commit -m "your message"
git push -u origin main
```